Tao Xu (徐韬)

Academic Curriculumn vitae

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Research Interests

> *I am devoted to understanding, overcoming, and exploiting uncertainties in dynamical systems.*

My research lies at the intersection of control, game theory, optimization, and learning. Particular foci are on probabilistic prediction of stochastic dynamical systems, decision-aware (predict-then-optimize) learning, and mixed-strategy differential games. These ideas anchor in networked and multi-agent systems, where they have found their way into industrial applications.

Education

2022–current Ph.D. Candidate in Control Science and Engineering at Shanghai Jiao Tong University
Zhiyuan Honor Program
Advisor: Jianping He

2018–2022 B.S. in Mathematics at Shanghai Jiao Tong University
Wu Wenjun Class
Thesis: Continuous-time Model of Stochastic Variance Reduced Gradient
Advisor: Jingwei Liang and Tao Luo

Publications

Journal Papers

2026 1. **Tao Xu**, Wang Xi, and Jianping He, Near-Optimal Mixed Strategy for Zero-Sum Linear-Quadratic Differential Games, IEEE Transactions on Automatic Control, 2026. (Full Paper) [doi]

2026 2. **Tao Xu** and Jianping He, Distributionally Robust Probabilistic Prediction for Stochastic Dynamical Systems, IEEE Transactions on Automatic Control, 2026. (Full Paper) [doi]

2025 3. **Tao Xu**, Yushan Li, and Jianping He, Probabilistic Predictability of Stochastic Dynamical Systems, Automatica, 2025. [doi]

Conference Papers

2026 4. Wenxiao Du, **Tao Xu**, Xiaoyu Luo, Chongrong Fang, Jianping He, Decision-Aware Learning for Context-Dependent LQR: A Smart Predict-then-Optimize Approach, 2026 IEEE 65th Conference on Decision and Control (CDC), 2026. (To Appear)

2026 5. Ling Yao, Xiangyun Rao, **Tao Xu**, Pangkit Fong, Jianping He, High-Efficiency Distributed Nonconvex Optimization Design with Certified Stopping Rule, 2026 IEEE 65th Conference on Decision and Control (CDC), 2026. (To Appear)

2023 6. **Tao Xu**, Wang Xi, and Jianping He, Differential Game with Mixed Strategies: A Weak Approximation Approach, 2023 IEEE 62nd Conference on Decision and Control (CDC), 2023. [doi]

2023 7. Yushan Li, **Tao Xu**, Jianping He, Cailian Chen, and Xinping Guan, Inferring State-Feedback Cooperative Control of Networked Dynamical Systems, IFAC-PapersOnLine, 2023. [doi]

2022 8. **Tao Xu** and Jianping He, Predictability of Stochastic Dynamical System: A Probabilistic Perspective, 2022 IEEE 61st Conference on Decision and Control (CDC), 2022. [doi]

Preprints

2026 9. **Tao Xu**, Wang Xi, and Jianping He, Near-Optimal Mixed Strategy for Zero-Sum Differential Games, arXiv:2308.01144, 2026.

2025 10. Xuyang Wang, Xinzhe Zhou, **Tao Xu**, Xiaoming Duan, Jianping He, A General and Efficient SE (3)-Equivariant Graph Framework: Encoding Symmetries with Complete Differential Invariants and Frames, 2025. [link]

2024 11. Jixian Liu, **Tao Xu**, Jianping He, Chongrong Fang, Smart Predict-then-Optimize Method with Dependent Data: Risk Bounds and Calibration of Autoregression, arXiv:2411.12653, 2024.

Software

2026 **ppsds**. Lead developer and maintainer. Open-source Python library for probabilistic prediction of stochastic dynamical systems: a predictability ceiling for systems known exactly, and a distributionally robust predictor with a worst-case floor over a declared ambiguity set. MIT; v0.0.3-alpha on PyPI. `https://github.com/Xuzerken/ppsds`

Funding

Jan'26–Dec'27 • NSFC, Young Student Basic Research Program (Doctoral Candidate), Weak Approximation Theory and Near-Optimal Algorithm Design for Mixed Strategies in Differential Games (Grant No. 625B2117), **Principal Investigator**.

Industry Collaboration

Sep'25–Dec'26 • Distributed Resource Allocation Networks. **Technical Lead**. Industry research collaboration with Huawei Technologies on distributed resource allocation over networks.

Honors & Awards

2026 • National Scholarship for Graduate Students.

2022–2024 • Rongchang Technology Innovation Scholarship, awarded by Shanghai Rongchang Public Welfare Foundation (~10 Recipients University-wide).

2022 • Outstanding Graduate of Shanghai Jiao Tong University.

2021 • Meritorious Winner, Mathematical Contest in Modeling (MCM) (Top 7%).

2020 • First Prize at the Shanghai Regional Level, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM) (Top 5%).

2020 • Second Prize at the Shanghai Regional Level (Category A, Mathematics Major), The Chinese Mathematics Competitions (CMC) (Top 10%).

2020 • Global Finalist, HackAsia Hackathon (Hong Kong).

2019 • Third Prize at the National Finals, China Society for Industrial and Applied Mathematics (CSIAM) Shenzhen Cup Mathematical Modeling Challenge (Top 0.2%, Only 3 Winners Per Track).

2017 • First Prize at the Provincial Level (Hubei Division), Chinese Mathematical Olympiad (CMO) - High School League (Top 0.5%, ~70 Winners Province-wide).

Teaching

2023–current Modern Control Theory, Teaching Assistant, Shanghai Jiao Tong University.

Service

2022-current Reviewer for IEEE Transactions on Automatic Control (TAC) ◦ Automatica ◦ IEEE Transactions on Control of Network Systems (TCNS) ◦ IEEE Transactions on Vehicular Technology (TVT) ◦ American Control Conference (ACC) ◦ IEEE Conference on Decision and Control (CDC) ◦ IFAC World Congress ◦ Learning for Dynamics and Control Conference (L4DC).

Apr'2026 Organizer for 2026 Discover Intelligence Via Engineering (DIVE) Forum. [link]